

KELIBIA, Tunisia

WMO: 607200

Lat: 36.850N

Lon: 11.083E

Elev: 98

StdP: 14.64

Time Zone: 1.00 (E01)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	42.9	44.7	33.3	28.0	49.0	35.8	30.9	49.8	22.3	54.1	19.2	54.5	3.8	300	0.352

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	13.6	91.4	74.5	89.2	74.7	87.1	74.7	79.5	85.4	78.4	84.1	77.4	83.1	6.8	310

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
77.7	144.7	83.2	76.8	140.3	82.2	75.5	133.9	81.2	43.1	85.8	41.9	84.1	40.8	83.1	83.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 17.4	15.2	13.5	DB	39.3	96.8	2.4	4.2	37.5	99.8	36.1	102.3	34.8	104.6	33.0	107.7	(4)
(5)			WB	37.0	81.0	2.1	1.4	35.5	81.9	34.3	82.7	33.2	83.5	31.7	84.5	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	66.1	54.4	54.3	57.0	60.8	66.8	73.9	79.1	80.6	75.8	70.4	62.5	56.8	(6)	
	DBStd	10.04	3.68	3.77	3.93	3.55	3.58	4.22	3.18	3.06	3.71	4.05	4.36	3.87	(7)	
	HDD50	20	8	8	2	0	0	0	0	0	0	0	0	2	(8)	
	HDD65	1395	329	300	250	133	22	1	0	0	0	6	99	255	(9)	
	CDD50	5892	144	128	219	323	519	717	903	947	773	632	374	213	(10)	
	CDD65	1792	0	0	2	6	76	268	438	482	323	173	23	1	(11)	
	CDH74	13529	0	0	22	38	210	1684	3994	4804	2201	559	17	0	(12)	
(13)	CDH80	4013	0	0	2	6	21	415	1353	1735	446	35	0	0	(13)	
(14)	Wind	WSAvg	6.1	6.0	6.2	6.5	6.7	6.8	6.4	6.6	5.7	5.6	5.2	5.6	5.8	(14)
(15) Precipitation	PrecAvg	23.0	3.3	2.7	2.1	1.5	0.8	0.3	0.2	0.3	2.0	3.1	3.1	3.6	(15)	
	PrecMax	37.2	7.9	8.7	4.7	3.8	2.4	1.7	2.0	1.3	7.6	8.9	9.9	7.8	(16)	
	PrecMin	14.5	0.5	0.0	0.1	0.0	0.0	0.0	0.0	0.0	0.1	0.4	0.5	0.4	(17)	
	PrecStd	5.5	2.1	2.0	1.4	1.0	0.7	0.4	0.4	0.4	1.5	1.9	1.8	1.9	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	67.6	68.3	77.1	80.2	82.6	91.4	96.1	96.4	89.5	82.8	75.7	69.5	(19)	
		MCWB	58.7	58.9	67.4	70.1	67.2	70.5	72.9	73.6	76.4	72.7	67.1	60.8	(20)	
	2%	DB	64.3	64.7	68.6	72.0	78.7	87.2	91.1	91.7	86.3	80.6	73.6	66.6	(21)	
		MCWB	57.6	56.6	58.7	62.1	65.6	71.7	73.9	75.4	75.2	72.8	66.3	60.2	(22)	
	5%	DB	62.3	62.6	66.0	69.5	75.9	84.4	88.2	89.6	84.3	78.9	71.7	65.0	(23)	
		MCWB	56.7	56.3	58.1	60.9	65.6	70.9	73.6	76.0	74.3	71.5	64.8	59.3	(24)	
	10%	DB	60.6	60.8	63.8	67.2	73.7	82.2	86.3	87.7	82.5	77.1	69.9	63.7	(25)	
		MCWB	55.6	55.2	57.1	59.9	64.6	70.2	73.8	76.2	73.4	70.3	64.0	58.3	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	60.3	60.2	69.1	70.8	71.6	76.5	80.5	81.0	79.9	76.3	70.4	64.0	(27)	
		MCDB	64.6	65.5	76.7	78.9	77.7	85.4	87.1	87.4	85.1	79.9	73.6	66.5	(28)	
	2%	WB	59.0	58.4	60.7	64.0	68.8	74.6	78.4	80.0	78.5	74.7	67.9	62.0	(29)	
		MCDB	62.6	62.7	66.3	69.3	74.4	82.1	84.5	86.1	83.1	78.4	71.8	65.3	(30)	
	5%	WB	57.6	57.1	59.2	62.4	67.3	73.2	77.2	79.0	77.2	73.3	66.4	60.2	(31)	
		MCDB	61.3	61.3	64.0	67.4	73.3	80.5	83.6	84.9	81.8	77.0	70.3	64.1	(32)	
	10%	WB	56.2	55.8	58.0	61.0	65.9	71.9	76.1	78.0	75.7	72.0	64.8	58.8	(33)	
		MCDB	59.9	59.9	62.4	65.4	72.0	78.9	82.6	83.9	80.5	75.9	69.3	63.0	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	11.3	11.8	13.0	12.6	12.7	13.7	14.1	13.6	12.2	11.7	11.6	11.4	(35)	
		MCDBR	13.5	13.9	16.9	16.2	15.4	16.7	17.2	16.2	13.1	12.6	13.2	13.2	(36)	
	5% WB	MCWBR	8.8	8.6	10.0	8.9	6.9	7.3	6.9	7.0	6.9	7.0	8.2	8.6	(37)	
		MCWBR	12.1	12.5	14.7	14.0	13.3	14.7	14.2	13.3	11.1	11.1	11.4	11.4	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.353	0.373	0.423	0.464	0.477	0.446	0.465	0.459	0.459	0.431	0.389	0.359	(40)	
	taud		2.442	2.371	2.213	2.118	2.122	2.255	2.207	2.236	2.237	2.273	2.353	2.421	(41)	
	Ebn at Noon		263	271	266	261	259	266	260	259	252	247	246	252	(42)	
	Edh at Noon		28	33	42	49	49	43	45	43	41	36	30	26	(43)	
(44) All-Sky Solar Radiation	RadAvg		746	1011	1414	1798	2126	2370	2389	2104	1583	1195	832	670	(44)	
(45)	RadStd		55	74	103	103	111	90	58	81	72	71	42	55	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	+0.68	N/A	N/A	+1.53	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (8 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page